

Aaron Xiong

(734)-730-8863 | aaronx@umich.edu | LinkedIn: aaron-xiong-um | Website: aaronxiong.com

EDUCATION

University of Michigan

Bachelor of Science in Engineering in Computer Science, Minor in Electrical Engineering

Cumulative GPA: 3.92 /4 (James B. Angell Scholar)

Courses: Computer Architecture, Machine Learning, Embedded Systems, Computer Security, Data Structures and Algorithms.

Ann Arbor, MI, USA

Aug. 2024 – May. 2027

(Expected)

SKILLS

- **Programming Languages:** C/C++, Python, HTML/CSS, JavaScript, MATLAB, System Verilog, ARMv7, Arduino, SQL
- **Libraries & Frameworks:** ReactJS, Flask, Tailwind CSS, Bootstrap, PyTorch, NumPy, Matplotlib, GTK, Libcurl
- **Tools:** Linux Ubuntu, Git, Docker, Oracle, VMWare, GDB, Wireshark, Ghidra, Verdi, AWS
- **Interests:** Dancing, Swimming, Snowboarding, Badminton, Tennis, Singing

WORK EXPERIENCE

Full-stack Developer for U-M School of Nursing

Jan. 2025 – Dec. 2025

- Designed a HIPAA-compliant, mobile-friendly, Full-stack web platform with a team of seven that aims to provide Nurse Health Coaching, Educational Content, and Personalized Community Resources for pregnant people and infant caregivers.
- Integrated Zoom and Qualtrics for nurse coaching and patient assessments, Google Maps for geospatial analysis, **Oracle Database** for patient data storage, and U-M's text-based generative AI tool Maizey for summarization functionalities.
- Engineered and optimized prompts for Maizey to accurately summarize meeting notes from nurse-patient Zoom sessions, generate personalized care plans for clients, and extract key needs from health surveys while mitigating hallucinations to ensure safety.
- Designed an accessible and intuitive UI with **Figma** through interviews and usability testing, ensuring a seamless user experience.

PROJECT EXPERIENCE

Voxband – the ISC Lab

Jan. 2026 – Present

- Engineered a wearable wristband integrating VPUs and air microphones to capture and process **body-conducted speech signals**.
- Designed, manufactured, and assembled **custom multi-layer PCBs** and 3D-printed enclosures to deploy the experimental prototype.
- Programmed firmware to support **4-channel USB audio streaming**, Wi-Fi communication, and external memory management.
- Built signal-processing and evaluation pipelines to benchmark **speaker diarization** robustness under diverse acoustic conditions.

Founder of RISC-M

Apr. 2026 – Present

- Founded the University of Michigan's first student organization dedicated to computer architecture, establishing a community of **25+ students** to engage in open-source projects, research discussions, and architecture design competitions.
- Defined the organization's operational roadmap and managed administrative coordination, communications, and logistics.
- Initiated and led a paper reading group, collaborating with faculty to curate and present foundational computer architecture research.
- Coordinated guest lectures and networking events, inviting industry engineers to give technical talks and share career insights.

Superscalar Processor Design (*SystemVerilog*)

Mar. 2026 – Apr. 2026

- Designed a 2-way superscalar, 4-stage, out-of-order RISC processor for the LC2K ISA and verified pipeline execution in **Verdi**.
- Implemented a parameterizable **micro-op queue** with a tree priority selector for hazard-aware out-of-order dispatch scheduling.
- Developed a set-associative instruction cache alongside an area-efficient banked branch predictor and branch target buffer, successfully **reducing the average CPI from 2.06 to 1.60**.
- Evaluated architectural tradeoffs by benchmarking module sizes and constructing Pareto analyses for PPA optimization.
- Managed team integration and debugging workflows through milestone tracking, diagram-first design, and atomic commits.

Site-Invariant Brain Age Estimator (*ML, Python*)

Jan. 2026 – Apr. 2026

- Designed a site-invariant ML pipeline to predict chronological brain age from structural MRI scans across 64 scanner configurations.
- Implemented a **3D ResNet-18** backbone with a Gradient Reversal Layer for Domain-Adversarial Training (**DANN**), reducing cross-site generalization error **from 2.99 to 1.24 years** on held-out scanners.
- Stabilized adversarial training by integrating a spectrally-normalized discriminator, label smoothing, and feature noise injection.
- Improved Mean Absolute Error **from 4.9 to 3.8 years** by integrating heteroscedastic **deep ensembles** to quantify aleatoric and epistemic uncertainty and filter out-of-distribution samples.

PUBLICATIONS

- Zou, K., Xiong, A., Zhang, Y., Zhang, F., Ren, Y., Yang, J., Lee, A., Bhushan, S., & Wang, L. (2025). *LiveOIBench: Can Large Language Models Outperform Human Contestants in Informatics Olympiads?* Preprint, <https://arxiv.org/abs/2510.09595>